P-C-450b
NOVEMBER 8, 1960

SUPERSEDING
Int. Fed. Spec. P-C-00450a (GSA-FSS)
November 2, 1959 and
Fed. Spec. P-C-450
October 23, 1957

FEDERAL SPECIFICATION

CLEANING SOLUTION, PORCELAIN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 The cleaner covered by this specification removes by chemical action iron or rust stains, urine, and hard water deposits from sanitary ware porcelain (sanitary ware made with a pottery base and a glaze fired at cone 9-12) and from steel and cast iron sanitary ware covered with a coating of grade AA acid-resistant porcelain enamel. Use of the cleaner requires supervision to guard against personal injury and misapplication. Sanitary ware coated with nonacid resistant porcelain enamel should not be acid cleaned since severe etching can result which would increase maintenance.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

L-M-186 — Medicinal Products and Clinical Laboratory Reagents; General Specification for Containers (Packaging and Packing).

UU-T-116 — Tape, Paper, Gummed, Water-Resistant.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102 — Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-10225 — Packaging of Glass Containers, One-Gallon Capacity and Smaller, for other than Medicinal Products for Domestic Shipment.

MIL-B-26701 — Bottles, Screw Cap and Carboys, Polyethylene, Flexible, Plastic.

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military specifications and standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Governmental:

Federal Insecticide, Fungicide, and Rodenticide Act.

(Copies may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price may be obtained from the Superintendent of Documents.)

Code of Federal Regulations:

49 CFR 71–90 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the code of Federal Regulations (Revised 1956) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–90 (Rev. 1956).")

NonGovernmental:

Uniform Freight Classification Committee — Rule 41.

(Copies may be obtained from the Uniform Freight Classification Committee, 1 Park Avenue, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Materials.—The cleaner shall contain hydrochloric acid (prepared by diluting commercial muriatic acid with water) and any necessary wetting agents, inhibitors, and other desirable additives. The ingredient materials used in the manufacture of the cleaner shall be intimately assembled and processed and suitable for the purpose intended.

3.2 Hydrochloric acid content. — Hydrochloric acid (HCl) shall assay not less than 5 nor more than 8 percent by weight when tested as specified in 4.4.1.

3.3 Sediment.—Shall not exceed 0.1 percent by volume when tested as specified in 4.4.2.

3.4 Federal Insecticide, Fungicide and Rodenticide Act.—If the cleaner is also intended as a restroom deodorant and sanitizer, or as a germicide and its registration under the Federal Insecticide Act is required by the Pesticide Regulations Section of the Department of Agriculture, Washington 25, D. C., it shall be so registered.

3.5 Stain removal.—Rust stains shall be removed within 5 minutes by the cleaner according to the test specified in 4.4.3.

3.6 Etching of porcelain enamel.—When tested as specified in 4.4.4, the cleaner shall not etch porcelain.

3.7 Corrosion inhibition.—The cleaner shall be correctly inhibited to minimize metallic corrosion. When cleaner is tested as specified in 4.4.5, the weight loss of the metal shall not exceed 0.15 percent.

3.8 Workmanship.—The material shall be clean, uniform and shall be free of any defects which may impair its utility.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be

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kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.—A lot shall consist of all porcelain cleaner manufactured as one batch and offered for delivery at one time.

4.3 Sampling and inspection for acceptance. —Sampling and inspection shall be in accordance with the provisions set forth in Military Standard MIL–STD–105, except where otherwise indicated hereinafter.

4.3.1 *Inspection of the end item.*

4.3.1.1 *Examination of the end item.* — Examination of the end item shall be in accordance with the classification of defects and the acceptable quality levels (AQL's) set forth below; the lot size shall be expressed in units of filled containers.

4.3.1.1.1 *Examination of the filled container for defects in design, construction, appearance and shipment and special marking.* — The sample unit for this examination shall be one filled container.

Examine	Defects
Container	Not type specified.
Condition of container	Any evidence of leakage.
Marking for shipment	Not in accordance with MIL–STD–129 or Fed. Std. No. 123 as applicable.
	Illegible, incorrect or incomplete.
Labeling	"Directions for Use" data missing, illegible, incorrect or incomplete.

4.3.1.1.2 *Examination of the finished product for workmanship.* — The sample unit for this examination shall be one filled container.

Examine	Defects
Workmanship	Not clean.
	Not uniform.
	Evidence of foreign matter.

4.3.1.1.3 *Examination of the filled containers for determining compliance with the requirements for net contents.*—The average net contents shall be not less than specified. The sample unit for this examination shall be one filled container.

4.3.1.1.4 *Inspection levels and AQL's for examinations.* — The inspection levels and the acceptable quality levels (AQL's) expressed in defects per 100 units as applicable shall be as follows:

	Inspection level	*AQL*
For examination in 4.3.1.1.1	L–8 of MIL–STD–105	0.10
For examination in 4.3.1.1.2	L–1 of MIL–STD–105	2.5
For examination in 4.3.1.1.3	L–2 of MIL–STD–105	N/A

4.3.2 *Testing of the end item.* — The end item shall be tested for the characteristics listed in table I. Determinations shall be made for each of the characteristics from each lot presented for examination, with no evidence of failure to meet unit requirements. All requirements are applicable to sample unit. The sample unit shall be 1-quart or 2-pints. The inspection level shall be L–2 of Military Standard MIL–STD–105.

4.4 Test procedures.

4.4.1 *Hydrochloric acid content.* — Weigh a glass-stoppered Erlenmeyer flask containing about 30 ml. of distilled water. Add rapidly 3 to 4 ml. of the thoroughly mixed sample, stopper and reweigh. Dilute to 50 ml. with distilled water and titrate with 1 *N* sodium hydroxide using methyl orange in-

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TABLE I.—*Testing of the end item*

Characteristic	Reference to requirement paragraph	Test method	Results reported as	
			Pass or fail	Numerically to nearest
Hydrochloric acid content	3.2	4.4.1	--	0.1 percent
Sediment	3.3	4.4.2	--	0.01 percent
Stain removal	3.5	4.4.3	--	second
Etching of porcelain enamel	3.6	4.4.4	X	------
Corrosion inhibition	3.7	4.4.5	--	0.001 percent

dicator. Calculate the percentage of HCl by employing the formula:

$$\frac{A \times N \times 0.03647 \times 100}{B}$$

where:

A = Milliliters of sodium hydroxide solution

N = Normality of sodium hydroxide solution

B = Weight, in grams, of the specimen.

4.4.2 *Sediment.* — Thoroughly mix the sample and take 100 milliliters and place in a graduated ASTM water and sediment tube. Centrifuge at an equivalent centrifugal force of 1500 revolutions per minute, and a diameter of swing (tip to tip of whirling tubes) of 16 inches for 10 minutes. Read the volume in the bottom of the tube.

Note.—If the available centrifuge has a swing varying from that stated above, calculate the proper speed using the following formula, where D represents the diameter of swing:

$$\text{r.p.m.} = 1500\sqrt{16/D}$$

4.4.3 *Stain removal.*

4.4.3.1 *Preparation of stained specimen.* — An unglazed porcelain streak plate (Arthur H. Thomas Catalog No. 8295 or equivalent type) shall be used. Prepare a 5 percent solution of ferric chloride, anhydrous, A.C.S. grade, with distilled water. Wet the plate on one side with ferric chloride solution and set the stain by baking the plate for 2 hours at 180°C.

4.4.3.2 *Procedure.* — Saturate a cloth or sponge with cleaner and apply to a stained plate prepared as specified in 4.4.3.1. Allow cleaner to react for 5 minutes and then note the color of the plate. The original white appearance of the plate should be restored.

4.4.4 *Etching of porcelain enamel.* — For this test, new 3-inch by 3-inch white, porcelain enameled steel plates[1] will be needed. The porcelain enamel shall be equivalent to a regular sanitary ware acid resistant enamel, and the enameling iron shall be 20 gauge.

4.4.4.1 *Procedure.* — Place approximately one gram of diatomaceous earth (Dicalite, Johns-Manville Hyflo or equivalent) on a plate to make a 5/8 inch diameter circle with a depth of 3/8 inch, and saturate the earth with 2 ml. of the cleaner. Cover the wetted area with a watch glass and set aside for 16 hours. At the expiration of 16 hours, wash with a rag or sponge and dry the plate and examine with a 5x magnifying glass for any marked degree of etching by comparing with a new plate under good lighting conditions

[1] Plates from Mr. Earl Skillicorn, Ferro Corporation, 4150 E. 56th Street, Cleveland 5, Ohio (or other equivalent sources) may be used.

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4.4.5 *Corrosion inhibition.*

4.4.5.1 *Preparation of coupons.* — Descale 2-inch by 2-inch by 0.035-inch new coupons cut from SAE 1010 hot rolled steel by immersing in concentrated reagent grade hydrochloric acid. The coupons shall then be rinsed, first with tap water, next with distilled water, and finally with reagent grade acetone. Dry the coupons at 105°C. for 15 minutes. Weigh the coupons to the nearest milligram and store in a desiccator. Proceed with the test in 4.4.5.2 with coupons prepared on the same day of the test.

4.4.5.2 *Procedure.* — Take 3 new coupons prepared as in 4.4.5.1 and place in 250-ml. beakers, one coupon to a beaker. Rest the coupon on a section of ⅛ inch glass rod bent into a V shape. Add 200 ml. of cleaner to each beaker. Allow the cleaner to remain unagitated in contact with the metal coupon for 24 hours and maintain uniform room temperature conditions. Remove the coupons after 24 hours, rinse two or three times with distilled water, and finally rinse with acetone, dry at 105°C. for 15 minutes, and weigh again. Determine the percent weight loss by the following formula and average the percentages:

$$\text{Percent weight loss} = \frac{\text{wt. before immersion} - \text{wt. after immersion}}{\text{wt. before immersion}} \times 100$$

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.* — The cleaner shall be packaged in 16-ounce or 32-ounce narrow-mouth glass bottles conforming to type I, class B of Federal Specification U–M–186 or in polyethylene bottles conforming to Military Specification MIL–B–26701. The bottles shall have closures as specified in Federal Specification U–M–186 or in Military Specification MIL–B–26701, as applicable. The closures shall be tightened to prevent leakage, and shall be fastened to the bottles with cel-o-seal, viscose, or other suitable material in accordance with Federal Specification U–M–186. The bottles in the quantity and arrangement prescribed in Federal Specification U–M–186 shall then be packaged in a style RSC W5c fiberboard box conforming to Federal Specification PPP–B–636. The bottles shall be separated by interlocking partitions made of W5c or W6c fiberboard conforming to Federal Specification PPP–B–636. The partitions shall form a snug-fitting cell for each bottle. A liner, with horizontal corrugations, of one piece construction, and extending the full depth of the box, shall cover the sides and ends of the box. Each box shall be provided between the inside faces of the liner with top and bottom pads, and with a pad between the layers of bottles in the multiple tier boxes. The liners and pads shall be made of W5c or W6c fiberboard conforming to Federal Specification PPP–B–636. The box shall be closed with 3 inch wide water-resistant tape conforming to Federal Specification UU–T–116.

5.1.2 *Level B.* — Sixteen- or thirty-two ounce quantities of the cleaner shall be packaged in the bottles and provided with the closures specified in 5.1.1.

5.1.3 *Level C.* — The cleaner shall be packaged in 16-ounce or 32-ounce bottles in accordance with the manufacturer's commercial practice, when adequate.

5.2 Packing.

5.2.1 *Level A.* — The cleaner, packaged in accordance with 5.1.1 shall be packed in a style 2, 2½, or 4 nailed wood box in conformance with class 2 of Federal Specification PPP–B–621, and the box shall be strapped in accordance with appendix thereto.

5.2.2 *Level B.* — The cleaner packaged as specified in 5.1.2, shall be packed in accord-

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ance with Military Specification MIL–P–10225.

5.2.3 *Level C.* — The cleaner packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall be in compliance with Rule 41, of the Uniform Freight Classification or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* — In addition to any special marking required by the contract or order, the shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* — In addition to 5.3.4 and any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

5.3.3 *Special marking.* — If any special marking is required on shipping containers, the marking shall comply with Interstate Commerce Commission Regulations for Transportation of Explosives and other Dangerous Articles by Land and Water, 49 CFR 71–90.

5.3.4 *Directions for use.* — Each unit package shall bear clear and concise directions for use. Since supervision is required in the use of this product, adequate caution notice shall be included on the label.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Storage surveillance. — Cleaner packaged in accordance with level B requirements must be inspected to determine condition when not used within three years. Cleaner not used within three years must either be replaced in accordance with level B requirements in this specification or with level A requirements if storage beyond an additional year is anticipated.

6.3 Transportation description.—Transportation description and minimum weights applicable to this commodity are:

Rail:

Cleaning Compounds, not otherwise indexed by name, liquid.

Carload minimum weight 86,000 pounds.

Motor:

Cleaning Compounds, not otherwise indexed, liquid.

Truckload minimum weight (W) 36.6 pounds, (W) subject to Rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—C E O Q.

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